

Learning Activity Operating Systems

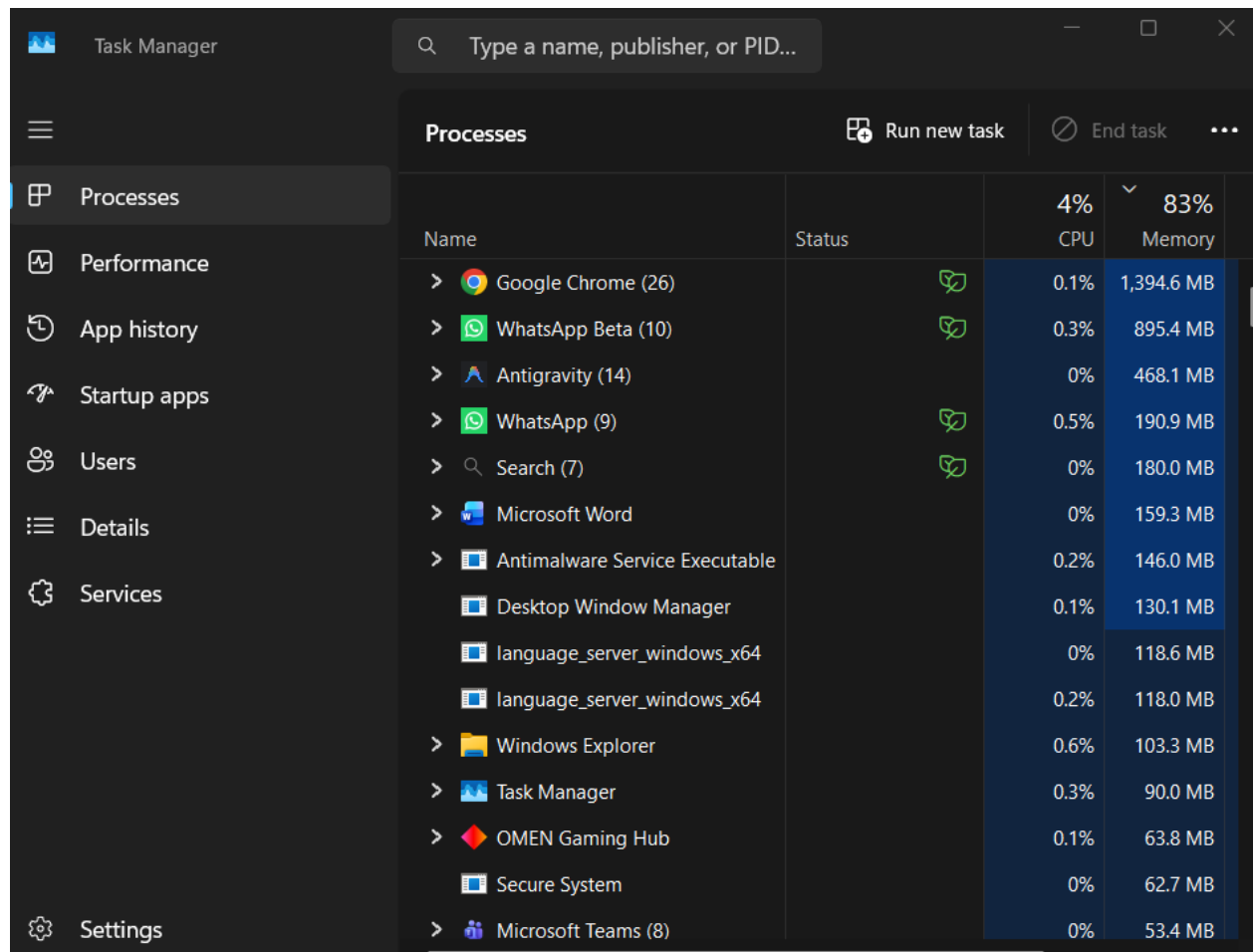
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Admission Number: I24AI012

- 1) Observation: Process Behavior Watch: Open Task Manager, Make observation of processes and their details. Write down all your observations.

In order to analyze how the processes behave in real time, I found opening up my Task Manager on my machine and observing various running process along with their properties.

So, when I opened Task Manager for the first time I saw that there were many more processes running than the applications in which I had manually opened. That made me realize there are tons of background processes running in the operating system which user cannot see directly.



The screenshot shows the Windows Task Manager application. The 'Processes' tab is selected, displaying a list of running processes. The left sidebar contains navigation options: Processes, Performance, App history, Startup apps, Users, Details, Services, and Settings. The top of the window has a search bar and buttons for 'Run new task', 'End task', and a menu icon. The process list includes columns for Name, Status, CPU usage, and Memory usage. The total system usage is shown as 4% CPU and 83% Memory.

Name	Status	CPU	Memory
Google Chrome (26)	Running	0.1%	1,394.6 MB
WhatsApp Beta (10)	Running	0.3%	895.4 MB
Antigravity (14)	Running	0%	468.1 MB
WhatsApp (9)	Running	0.5%	190.9 MB
Search (7)	Running	0%	180.0 MB
Microsoft Word	Running	0%	159.3 MB
Antimalware Service Executable	Running	0.2%	146.0 MB
Desktop Window Manager	Running	0.1%	130.1 MB
language_server_windows_x64	Running	0%	118.6 MB
language_server_windows_x64	Running	0.2%	118.0 MB
Windows Explorer	Running	0.6%	103.3 MB
Task Manager	Running	0.3%	90.0 MB
OMEN Gaming Hub	Running	0.1%	63.8 MB
Secure System	Running	0%	62.7 MB
Microsoft Teams (8)	Running	0%	53.4 MB

When I clicked the Processes tab, processes were classified based on Apps, Background processes, and Windows processes. Under Apps, applications such as a browser and file explorer showed up along with many unknown processes running in the background.

This contains details like CPU usage, Memory usage, Disk activity & Network usage for each process. And browsing processes (especially with multiple tabs open) used up plenty of CPU/memory.

Now we found one of them interesting, CPU usage was keep changing. On an idle system, some background processes were still consuming small amounts of CPU. This indicates that the operating system is constantly working in the background.

I noticed that the memory usage was also pretty much if not high with just a few applications open. The reason is likely that background services and system processes take up memory for the continuous operation of your system.

Another thing I noticed was that some processes had multiple instances (for example, browser processes). This indicates that modern applications use multiple processes for better performance and stability.

Task Manager				
Type a name, publisher, or PID...				
Processes				
Run new task End task ...				
		4%	83%	
Name	Status	CPU	Memory	
Google Chrome (24)		0.3%	1,376.8 MB	
Google Chrome		0%	223.9 MB	
Google Chrome	Efficiency ...	0%	194.5 MB	
Google Chrome		0%	187.0 MB	
Google Chrome	Efficiency ...	0%	169.2 MB	
Google Chrome	Efficiency ...	0%	124.8 MB	
Google Chrome		0.3%	99.0 MB	
Google Chrome	Efficiency ...	0%	91.7 MB	
Google Chrome	Efficiency ...	0%	45.9 MB	
Google Chrome	Efficiency ...	0%	36.4 MB	
Google Chrome	Efficiency ...	0%	34.6 MB	
Google Chrome	Efficiency ...	0%	32.3 MB	
Google Chrome	Efficiency ...	0%	28.3 MB	
Google Chrome		0%	21.1 MB	
Google Chrome	Efficiency ...	0%	17.2 MB	

I was able to view more details like Process ID (PID), Status, and Priority when I went to the Details tab. Each Process had its own PID, which makes it easier for the operating system to distinguish between them.

Task Manager

Type a name, publisher, or PID...

Processes

Performance

App history

Startup apps

Users

Details

Services

Settings

Details

Run new task

End task

Name	PID	Status	User name	CPU	Worki...	Platform	UAC \
AggregatorHost.exe	8824	Runni...	SYSTEM	00	0 K	64 bit	Not a
ai.exe	23788	Runni...	Suraj Shah	00	0 K	64 bit	Disab
aimgr.exe	6840	Runni...	Suraj Shah	00	0 K	64 bit	Disab
Antigravity.exe	23404	Runni...	Suraj Shah	01	-32 K	64 bit	Disab
Antigravity.exe	18064	Runni...	Suraj Shah	00	0 K	64 bit	Disab
Antigravity.exe	22836	Runni...	Suraj Shah	00	0 K	64 bit	Disab
Antigravity.exe	7684	Runni...	Suraj Shah	00	0 K	64 bit	Disab
Antigravity.exe	23576	Runni...	Suraj Shah	00	96 K	64 bit	Disab
Antigravity.exe	23756	Runni...	Suraj Shah	00	0 K	64 bit	Disab
Antigravity.exe	24060	Runni...	Suraj Shah	00	0 K	64 bit	Disab
Antigravity.exe	24068	Runni...	Suraj Shah	00	0 K	64 bit	Disab
Antigravity.exe	24164	Runni...	Suraj Shah	00	8 K	64 bit	Disab
Antigravity.exe	24468	Runni...	Suraj Shah	00	0 K	64 bit	Disab
Antigravity.exe	24540	Runni...	Suraj Shah	00	0 K	64 bit	Disab
Antigravity.exe	16084	Runni...	Suraj Shah	00	0 K	64 bit	Disab
Antigravity.exe	10116	Runni...	Suraj Shah	00	0 K	64 bit	Disab
Antigravity.exe	23348	Runni...	Suraj Shah	00	0 K	64 bit	Disab
AppActions.exe	20088	Runni...	Suraj Shah	00	0 K	64 bit	Disab
AppHelperCap.exe	3756	Runni...	SYSTEM	00	0 K	64 bit	Not a
ApplicationFrameHo...	16868	Runni...	Suraj Shah	00	0 K	64 bit	Disab
backgroundTaskHost...	22948	Runni...	Suraj Shah	00	0 K	64 bit	Disab
BridgeCommunicati...	16628	Runni...	Suraj Shah	00	0 K	64 bit	Disab

In general, this experiment allowed me to realize that multiple processes run concurrently on any computer at all times, and resource consumption is also variable. Furthermore, I have gained practical insight into how an operating system handles processes in real time.

- 2) Design Your Own Scheduling Algorithm subject to special requirements and specific environment. Check the applicability of scheduling algorithms in applications other than Operating systems.

Objectives

Objectives of this activity include:

- Developing a custom scheduling algorithm considering specific system requirements
- Analyzing its functionality
- Understanding the application of scheduling concepts in areas other than operating systems

Algorithm Suggested

A new scheduling algorithm was proposed by me named 'Adaptive Priority with Aging' (APA). This algorithm uses concepts of priority scheduling with aging to prevent starvation problems.

Workings of the Algorithm

1. A new process is characterized with:
 - o Initial priority
 - o Arrival time
 - o Burst time
2. Process is chosen as per:
 - o Highest priority
 - o When priorities are same, then burst time of shorter process gets preference
3. Aging technique:
 - o If the process is waiting for too long in the queue, then its priority increases
 - o It prevents starvation of low priority processes
4. Adjustment of priority:
 - o If the process uses the CPU too much continuously, then priority decreases slightly
 - o CPU is utilized in a fair manner

Requirement & Working Environment

This type of algorithm would be useful in situations where:

- Fairness and response are important considerations
- Both short & long processes run simultaneously
- Multitasking environment exists

Observations

During designing the above-mentioned algorithm, I observed that:

- Simply relying on priority creates starvation problem
- Using aging solves the problem
- Combining two factors leads to better results than depending upon any one factor

Scheduling in Other Applications

Not only operating systems make use of scheduling techniques. Scheduling is also applicable in other areas:

1. Web Servers

Users' requests can be prioritized depending on their importance.

2. Cloud Computing

Tasks can be assigned to virtual machines depending on resources available at the moment.

3. Banking Systems

Transactions can be prioritized (important transactions, for example).

4. Hospital Management Systems

Patients can be scheduled according to the seriousness of their condition rather than arrival order.

5. Traffic Control Systems

Traffic signals can be adjusted depending on the level of traffic rather than set time intervals.

Problems Encountered

When designing an algorithm, one problem that I encountered was finding a balance between efficiency and fairness. Raising priority too fast could lead to poor system performance, but raising it too slowly could result in starvation.

Conclusion

This assignment allowed me to see that traditional scheduling algorithms are not enough for all situations. Custom algorithms such as APA can provide more opportunities for controlling system performance. Additionally, it allowed me to understand that scheduling is a universal technique that can be applied outside of operating systems.

- 3) Design Operating for Special Users such as Users in Critical Mission or cognitive, visual, or auditory disabilities

Objective

The objective of this assignment is:

- To identify the requirements of special users
- To implement certain features in OS supporting these users
- To enhance the usability, reliability, and accessibility

Design for Critical Mission Users

Special users working in critical mission applications (such as air traffic control or any other medical application) demand reliability and responsive system.

Important aspects in this regard may include:

- Real time processing capability

The system should process instantly without any delay

- Reliability and robustness of system

The system should work properly even in case of failure of certain hardware components

- Reducing user distractions

The interface should not contain unnecessary elements

- Priority task handling mechanism

Priority tasks have to be completed first

- System monitoring and alerts

Alerts should be raised instantly in case of system failure

Thinking along the above lines, I understood that a little delay in such systems would prove very disastrous.

Design for Visually Impaired Users

For visually impaired special users, voice interaction will be very useful.

This would include:

- Speech recognition and screen reader facility for reading the displayed text
- Voice command to navigate through the system
- High contrast theme for partially visually impaired people
- Shortcuts for keyboard operation without mouse usage

I realized during this exercise that even labeling the button with appropriate names plays a vital role here.

Operating System Design for Special User Groups

Operating System Design for Users with Auditory Disabilities

For users having difficulty in relying on sound:

- Visual alarms in place of auditory alarm
- Subtitle and captioned multimedia applications
- Vibrations or visual alarms (on supported systems)

This helps ensure that no important system message is overlooked.

Operating System Design for Users with Cognitive Disabilities

Cognitive disabilities make the users face problems in comprehending complicated system interface designs.

Features used are:

- Minimal and neat user interface
- Step-by-step guidance during the execution of tasks
- Usage of icons and images instead of texts
- Uniformity in design

While working on this part of the assignment, I realized that minimalistic interface benefits even non-disabled users.

Challenges Encountered

The first challenge was to strike a balance between simplicity and functional features. On one hand, simplicity reduces the number of features, whereas complexity would make the special users find it challenging to use.

Secondly, the requirements of different users vary, thus requiring the designer to keep in mind varied requirements.

Conclusion

Designing an operating system for special users involves many factors such as accessibility, reliability, and usability. Real-time processing, assistive technologies, and simplified interfaces are crucial in designing such systems.

- 4) How the threads synchronize with each other for applications.

Objective

The purpose of this exercise is:

- To gain insight into why thread synchronization is needed
- To learn about various techniques employed for synchronization
- To explore how synchronization is used in practical applications

Need for Thread Synchronization

During concurrent execution, multiple threads may attempt to read/write to the same data, which can lead to problems such as:

- Race condition – where the outcome depends on the order of operations
- Inconsistent state of shared variables
- Uncertain behavior

For instance, if two threads try to alter the same variable simultaneously, the end result might be incorrect.

Synchronization Techniques

Several synchronization techniques are commonly utilized:

1. Mutex/Mutual Exclusion Lock

Mutex guarantees that a particular segment of code is executed by only one thread at a time. The other threads have to wait until the lock is released.

2. Semaphore

Semaphore allows for controlling access to a resource by multiple threads simultaneously. It is beneficial when more than one copy of a resource exists.

3. Monitor

Monitor represents an advanced synchronization tool that combines mutex and condition variables.

Working Example

Suppose we have several threads that write into the same file. In such cases, if synchronization is not used, the data can become messed up.

When using a mutex,

- The thread acquires the file first
- The other threads will be waiting until it releases the file
- The thread then writes and releases the file

Thus, using a mutex will make sure that no more than one thread writes into the file at any point in time.

Applications of Thread Synchronization

Thread synchronization is required in several real-life situations:

- Banking system
- Database management systems
- Operating systems

- **Web servers**
- **Multiplayer games**

Challenges Faced

In my study about thread synchronization, I found that while synchronization helps to achieve correctness, it also has some drawbacks, such as reduced performance. Moreover, inappropriate use of locks can create problems, such as deadlocks, where the threads wait indefinitely.

Conclusion

Thread synchronization is very important in ensuring correct execution of multi-threaded applications. Methods of synchronization include mutex, semaphores, monitors, etc. This learning activity was quite helpful in realizing the significance of synchronization and its problems.

5) Try an Unfamiliar Operating Systems.

Objective

The objective of this activity includes:

- To work on a new operating system
- To understand the features and functionality of the operating system
- To compare this operating system with my usual operating system

Operating System Used

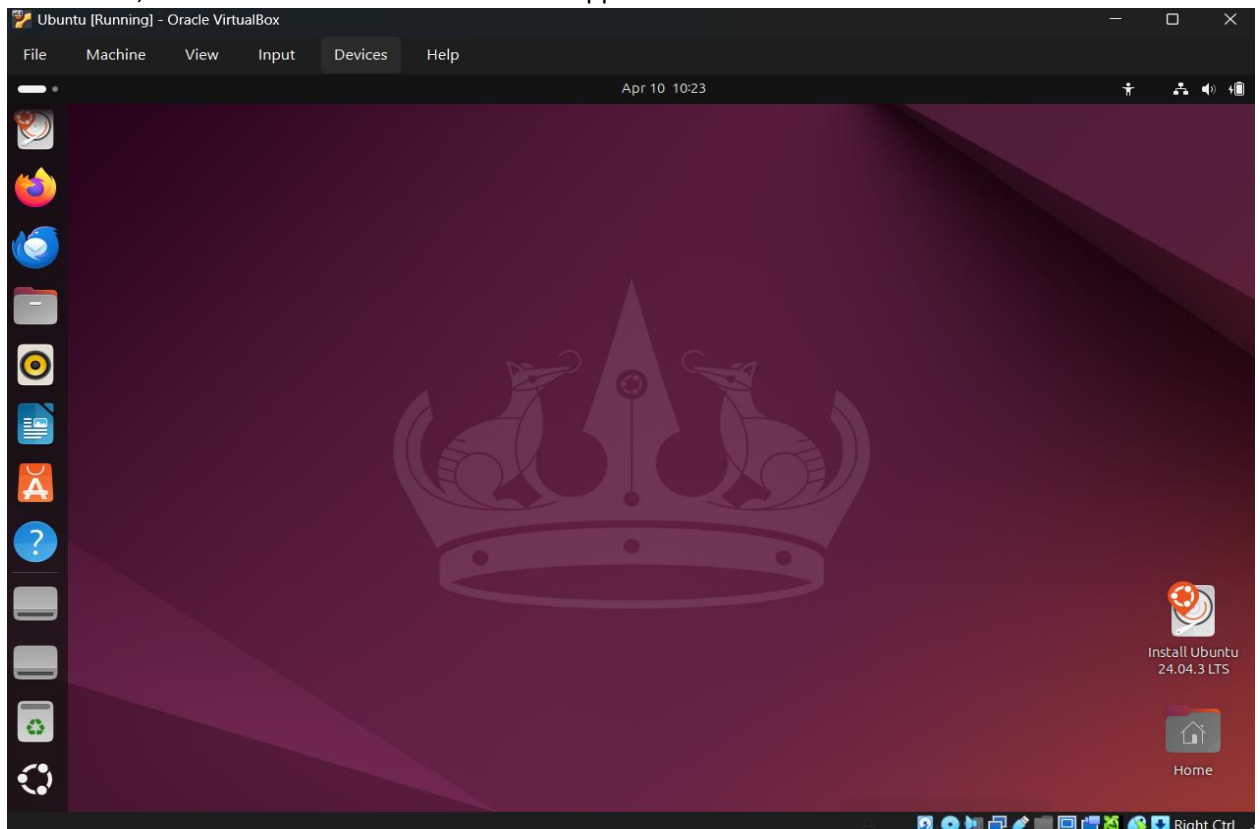
In this assignment, I experimented with the Ubuntu OS (Linux-based). In general, I had been working on the Windows Operating System.

Installation / Setup

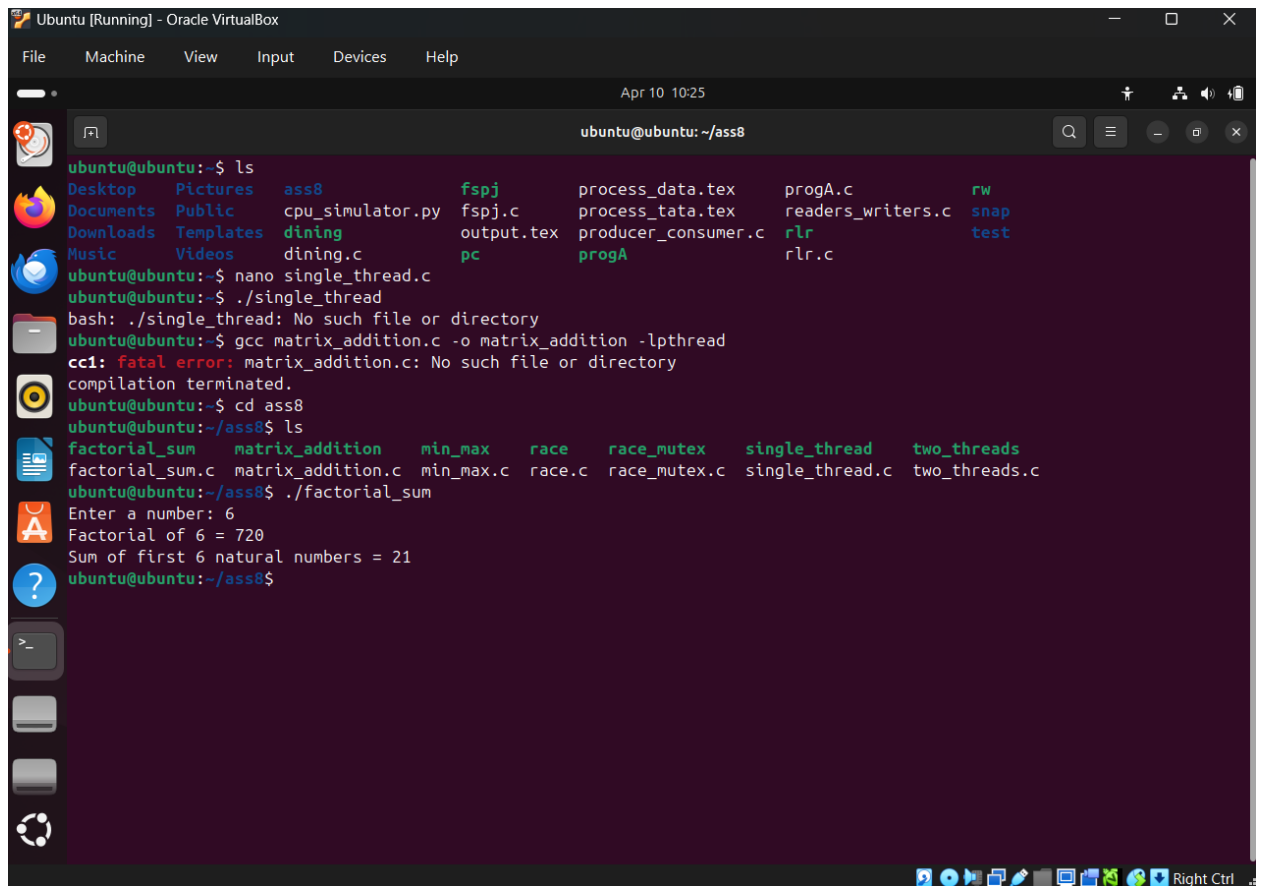
Ubuntu was installed on my system by setting up a virtual machine. Installation took some time and it was slightly difficult due to being different from other applications that were usually installed.

Observations

- The interface was completely new from that I was familiar with. Rather than having the start menu, there was the sidebar as well as an application launcher.



- I noticed that many tasks could be done using the terminal, which was new to me. At first, it felt complicated, but it also seemed powerful.

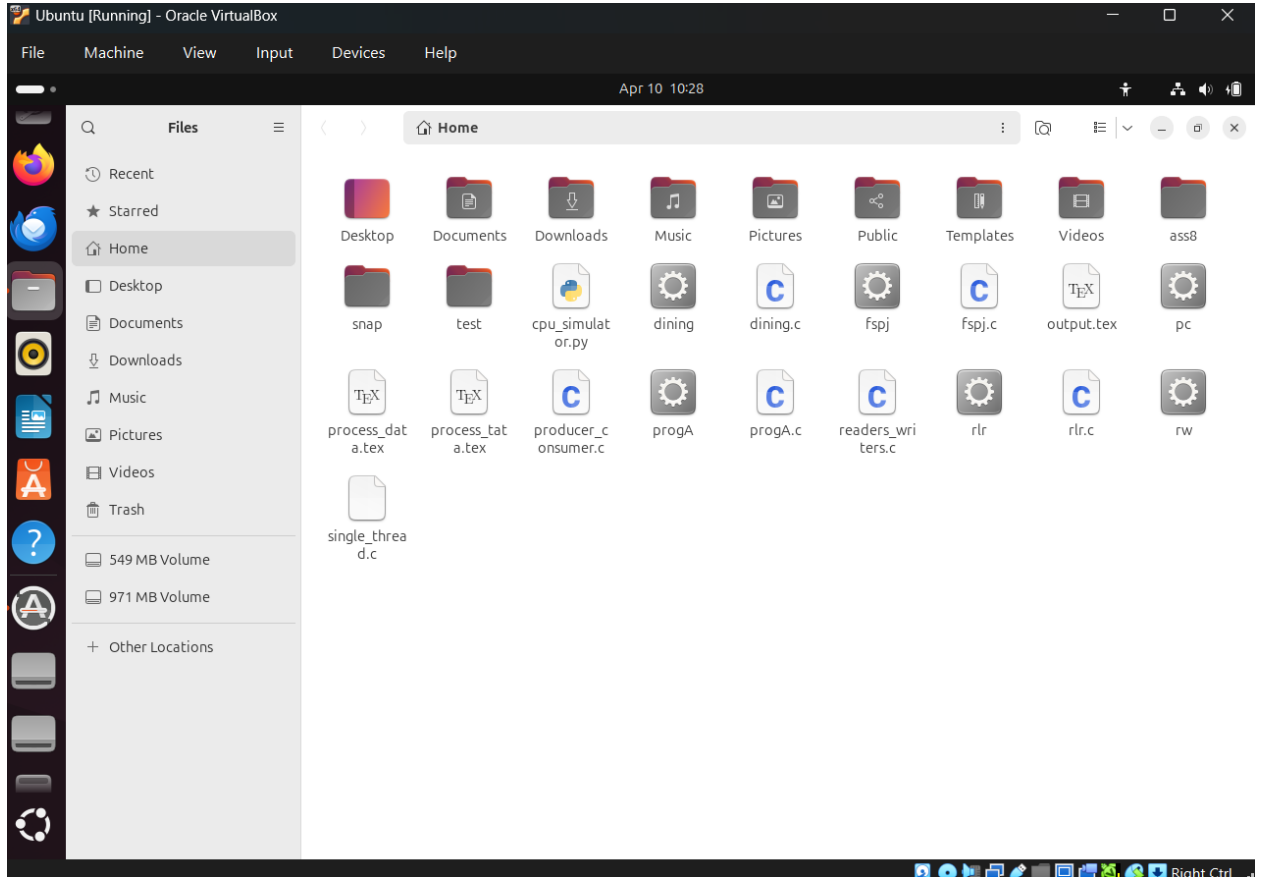


```
Ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Apr 10 10:25
ubuntu@ubuntu: ~/ass8
ubuntu@ubuntu:~$ ls
Desktop  Pictures  ass8      fspj      process_data.tex  progA.c      rw
Documents Public    cpu_simulator.py fspj.c    process_tata.tex  readers_writers.c  snap
Downloads Templates dining    output.tex  producer_consumer.c  rlr          test
Music    Videos  dining.c   pc         progA             rlr.c
ubuntu@ubuntu:~$ nano single_thread.c
ubuntu@ubuntu:~$ ./single_thread
bash: ./single_thread: No such file or directory
ubuntu@ubuntu:~$ gcc matrix_addition.c -o matrix_addition -lpthread
cc1: fatal error: matrix_addition.c: No such file or directory
compilation terminated.
ubuntu@ubuntu:~$ cd ass8
ubuntu@ubuntu:~/ass8$ ls
factorial_sum  matrix_addition  min_max  race  race_mutex  single_thread  two_threads
factorial_sum.c matrix_addition.c min_max.c race.c  race_mutex.c single_thread.c two_threads.c
ubuntu@ubuntu:~/ass8$ ./factorial_sum
Enter a number: 6
Factorial of 6 = 720
Sum of first 6 natural numbers = 21
ubuntu@ubuntu:~/ass8$
```

- The system felt lightweight and fast, especially compared to my usual OS.
- Software installation was different. Instead of downloading from websites, I could install applications using commands or a software manager.

```
Ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Apr 10 10:27
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ sudo apt install vlc
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  fonts-freefont-ttf i965-vd-driver intel-media-va-driver liba52-0.7.4 libaacs0 libaribb24-0t64 libass9 libavcodec60
  libavformat60 libavutil58 libbdplus0 libbluray2 libcdcb2 libchromaprint1 libbson1 libcodecs2-1.2 libdav1d7
  libdc1394-25 libdca0 libdouble-conversion3 libdrm-amdgpu1 libdrm-common libdrm-intel1 libdrm-nouveau2 libdrm-radeon1
  libdrm2 libdvbpsi10 libdvbnav4 libdvdrad8t64 libebml5 libegl-mesa0 libfaad2 libgbm1 libgl1-mesa-dri libglx-mesa0
  libgme0 libgsm1 libhwy1t64 libigdm12 libixml1t64 libjxl0.7 libkate1 libllvm20 liblua5.2-0 libmad0 libmatroska7
  libmbcrypto7t64 libmd4c0 libmpcdec6 libmpeg2-4 libmysofa1 libnorm1t64 libopenmpt-modplug1 libopenmpt0t64
  libpcre2-16-0 libpgm-5.3-0t64 libpostproc57 libprotobuf-lite32t64 libprotobuf32t64 libproxy-tools libqt5core5t64
  libqt5dbus5t64 libqt5gui5t64 libqt5network5t64 libqt5qml5 libqt5qmlmodels5 libqt5quick5 libqt5svg5
  libqt5waylandclients5 libqt5waylandcompositor5 libqt5widgets5t64 libqt5x11extras5 librabbitmq4 librav1e0
  libresid-builder0c2a librist4 libshine3 libsidplay2 libsnappy1v5 libsoxr0 libspatialaudio0t64 libstr1.5-gnutls
  libssh-4 libssh-gcrypt-4 libssh2-1t64 libsvtav1enc1d1 libswresample4 libswscale7 libudfread0 libunibreak5
  libupnp17t64 libva-drm2 libva-wayland2 libva-x11-2 libva2 libvdpau1 libvlc-bin libvlc5 libvlccore9 libvpl2
  libx264-164 libx265-199 libxcb-composite0 libxcb-xinerama0 libxcb-xinput0 libxvidcore4 libxmq5 libzvt-common
  libzvt0t64 mesa-libgallium mesa-va-drivers mesa-vdpau-drivers mesa-vulkan-drivers ocl-icd-libopencl1
  qt5-gtk-platformtheme qttranslations5-l10n qtwayland5 va-driver-all vdpau-driver-all vlc-bin vlc-data vlc-l10n
  vlc-plugin-access-extra vlc-plugin-base vlc-plugin-notify vlc-plugin-qt vlc-plugin-samba vlc-plugin-skins2
  vlc-plugin-video-output vlc-plugin-video-splitter vlc-plugin-visualization
Suggested packages:
  i965-vd-driver-shaders libcuda1 libnvcuvid1 libnvidia-encode1 libbluray-bdj libdvdcss2 qgnomeplatform-qt5
  qt5-image-formats-plugins qt5-qmltooling-plugins opencl-icd libvdpau-va-gli vlc-plugin-fluidsynth vlc-plugin-jack
  vlc-plugin-pipewire vlc-plugin-svg
The following NEW packages will be installed:
  fonts-freefont-ttf i965-vd-driver intel-media-va-driver liba52-0.7.4 libaacs0 libaribb24-0t64 libass9 libavcodec60
  libavformat60 libavutil58 libbdplus0 libbluray2 libcdcb2 libchromaprint1 libbson1 libcodecs2-1.2 libdav1d7
  libdc1394-25 libdca0 libdouble-conversion3 libdvbpsi10 libdvbnav4 libdvdrad8t64 libebml5 libfaad2 libgme0 libgsm1
  libhwy1t64 libigdm12 libixml1t64 libjxl0.7 libkate1 libllvm20 liblua5.2-0 libmad0 libmatroska7 libmbcrypto7t64
  libmd4c0 libmpcdec6 libmpeg2-4 libmysofa1 libnorm1t64 libopenmpt-modplug1 libopenmpt0t64 libpcre2-16-0
  libpgm-5.3-0t64 libpostproc57 libprotobuf-lite32t64 libproxy-tools libqt5core5t64 libqt5dbus5t64 libqt5gui5t64
  libqt5network5t64 libqt5qml5 libqt5qmlmodels5 libqt5quick5 libqt5svg5 libqt5waylandclients5 libqt5waylandcompositor5
  libqt5widgets5t64 libqt5x11extras5 librabbitmq4 librav1e0 libresid-builder0c2a librist4 libshine3 libsidplay2 libsnappy1v5 libsoxr0 libspatialaudio0t64 libstr1.5-gnutls
  libssh-4 libssh-gcrypt-4 libssh2-1t64 libsvtav1enc1d1 libswresample4 libswscale7 libudfread0 libunibreak5 libupnp17t64 libva-drm2 libva-wayland2 libva-x11-2 libva2 libvdpau1 libvlc-bin libvlc5 libvlccore9 libvpl2
  libx264-164 libx265-199 libxcb-composite0 libxcb-xinerama0 libxcb-xinput0 libxvidcore4 libxmq5 libzvt-common libzvt0t64 mesa-libgallium mesa-va-drivers mesa-vdpau-drivers mesa-vulkan-drivers ocl-icd-libopencl1
  qt5-gtk-platformtheme qttranslations5-l10n qtwayland5 va-driver-all vdpau-driver-all vlc-bin vlc-data vlc-l10n vlc-plugin-access-extra vlc-plugin-base vlc-plugin-notify vlc-plugin-qt vlc-plugin-samba vlc-plugin-skins2
  vlc-plugin-video-output vlc-plugin-video-splitter vlc-plugin-visualization
```

- File structure was also different, with directories like /home, /bin, etc., which I had not seen before.



Comparison with Familiar OS

Ubuntu compared with Windows:

- I gained more freedom with Ubuntu
- There was a need for knowledge about the terminal
- More secure than windows and less virus infected
- BUT there was lack of programs I usually work on

Problems Encountered

- In the beginning, I was unable to do things since everything looked new to me
- Terminal commands were hard to memorize
- It took time to find certain settings

Eventually, it got easier to work with it.

Conclusion

I think trying something unfamiliar like a new OS can be good practice. It made me realize that each OS is designed differently depending on what it aims to do.